

Variance and Standard Deviation

Introduction

Variance and **Standard Deviation** are measures of dispersion. They tell us how much the observations in a dataset differ or spread out from the mean.

- **Low variance / low standard deviation** → values are close to the mean and are more consistent.
- **High variance / high standard deviation** → values are more spread out and are more varied.

Population Variance Formula

$$\sigma^2 = \frac{\sum (x - \mu)^2}{N}$$

Where:

- x = individual observation
- μ = mean
- N = total number of observations
- σ^2 = variance

Population Standard Deviation Formula

$$\sigma = \sqrt{\frac{\sum (x - \mu)^2}{N}}$$

Standard deviation is simply the **square root of variance**.

Q1. Variance of Daily Step Counts

Question

Calculate the variance for daily step counts from a fitness tracker.

Example Data

Consider the following daily step counts:

4000, 5000, 6000, 7000, 8000

Step 1: Calculate the Mean

Mean=4000+5000+6000+7000+80005\text{Mean}= \frac{4000+5000+6000+7000+8000}{5} =300005=\frac{30000}{5} \mu=6000\boxed{\mu=6000}

Step 2: Calculate Deviations from the Mean

Steps xx	Mean μ	$x-\mu$	$(x-\mu)^2$
4,000	6,000	-2,000	4,000,000
5,000	6,000	-1,000	1,000,000
6,000	6,000	0	0
7,000	6,000	1,000	1,000,000
8,000	6,000	2,000	4,000,000
Total			10,000,000
			0

Step 3: Calculate Variance

$\sigma^2=10,000,0005\sigma^2= \frac{10,000,000}{5} \sigma^2=2,000,000\boxed{\sigma^2=2,000,000}$

Therefore, the **variance of the daily step counts is 2,000,000 steps²**.

Interpretation

The variance measures how widely the daily step counts are spread around the average of **6,000 steps**.

A larger variance indicates greater variation in daily activity.

Q2. Standard Deviation of Monthly Mobile Data Usage

Question

Calculate the standard deviation for monthly mobile data usage and decide whether the usage is **consistent** or **varied**.

Example Data

Suppose monthly mobile data usage is:

10, 11, 10, 12, 11, 10, 11, 10, 12, 11, 10

All values are in **GB**.

Step 1: Calculate the Mean

Mean = $\frac{10+11+10+12+11+10+11+10+12+11+10}{6} = \frac{64}{6} = 10.67$
 $\mu = 10.67$

Step 2: Calculate the Squared Deviations

Usage xx	$x - \mu$ - μ	$(x - \mu)^2$ - μ^2
10	-0.67	0.45
11	0.33	0.11
10	-0.67	0.45
12	1.33	1.77
11	0.33	0.11
10	-0.67	0.45
Total		3.34

Step 3: Calculate Variance

$\sigma^2 = \frac{3.34}{6} = 0.56$
 $\sigma^2 \approx 0.56$

Step 4: Calculate Standard Deviation

$\sigma = \sqrt{0.56} \approx 0.75$ GB

Interpretation

The standard deviation is approximately **0.75 GB**, which is relatively small compared with the average usage of about **10.67 GB**.

For this assignment, we can set the following simple threshold:

- Standard deviation < 2 GB → Low / Consistent usage
- Standard deviation ≥ 2 GB → High / Varied usage

Since:

$$0.75 < 20.75 < 2$$

the mobile data usage can be classified as:

Consistent\boxed{\text{Consistent}}

Explanation

The monthly usage values are close to one another, ranging only from **10 GB to 12 GB**. Therefore, the low standard deviation indicates that the user's mobile data consumption is relatively consistent.

Q3. Compare Variation in Coding Test and Music Quiz Scores

Consider the following marks.

Coding Test

60, 65, 70, 75, 8060,\ 65,\ 70,\ 75,\ 80

Music Quiz

40, 55, 70, 85, 10040,\ 55,\ 70,\ 85,\ 100

We will calculate the variance and standard deviation for both datasets.

A. Coding Test

Mean

$$\mu = \frac{60+65+70+75+80}{5} \quad \mu = \frac{350}{5}$$

$$\mu = 70 \boxed{\mu = 70}$$

Variance

Marks	Deviation	Squared Deviation
60	-10	100
65	-5	25

70	0	0
75	5	25
80	10	100
Total		250

$$\sigma^2 = \frac{250}{5} \text{ Variance} = 50 \boxed{\text{Variance} = 50}$$

Standard Deviation

$$\sigma = \sqrt{50} \approx 7.07 \boxed{\sigma \approx 7.07}$$

B. Music Quiz

Mean

$$\mu = \frac{40+55+70+85+100}{5} \mu = \frac{350}{5}$$

$$\mu = 70 \boxed{\mu = 70}$$

Variance

Marks	Deviation	Squared Deviation
40	-30	900
55	-15	225
70	0	0
85	15	225
100	30	900
Total		2,250

$$\sigma^2 = \frac{2250}{5} \text{ Variance} = 450 \boxed{\text{Variance} = 450}$$

Standard Deviation

$$\sigma = \sqrt{450} \approx 21.21 \boxed{\sigma \approx 21.21}$$

Comparison

Test	Mean	Variance	Standard Deviation
Coding Test	70	50	7.07

Music Quiz 70 450 21.21

Conclusion

The **Music Quiz has more variation** in scores because its variance (**450**) and standard deviation (**21.21**) are much higher than those of the Coding Test.

Although both tests have the same mean score of **70**, the music quiz scores are much more spread out from the mean.

Simple Interpretation

Coding Test: Scores are closer to 70 → **less variation**

Music Quiz: Scores are farther from 70 → **more variation**

Therefore, **standard deviation helps us understand the consistency or spread of scores even when two datasets have the same mean.**

Q4. Handling Empty or Single-Value Data

Concept

Variance and standard deviation measure **variation among observations**.

If there are:

- **No observations** → there is no data to measure.
- **Only one observation** → there is no variation because there is only one value.

Therefore, for this assignment, we can define:

Standard Deviation=0\boxed{\text{Standard Deviation}=0}

for an empty dataset or a dataset containing only one value.

Example 1: Empty Dataset

[]

There are no observations.

Therefore:

Standard Deviation=0\boxed{\text{Standard Deviation}=0}

Example 2: One Value

[50][50]

The only value is 50, and the mean is also 50.

Deviation:

$$50 - 50 = 0 \quad 50 - 50 = 0$$

Squared deviation:

$$0^2 = 0 \quad 0^2 = 0$$

Therefore:

$$\text{Standard Deviation} = 0 \quad \boxed{\text{Standard Deviation} = 0}$$

Example 3: Two or More Values

For example:

[40, 50, 60][40, \ 50, \ 60]

There is variation between the values, so the normal standard deviation formula can be applied.